Learning Chaotic Dynamics on the Lorenz System

Scientific Machine Learning Course Project

***Abstract*—We study forecasting and dynamical reconstruction on the chaotic Lorenz system using three learned models: an MLP one-step predictor, a residual discrete-time predictor, and a Neural ODE. The residual model achieves the best one-step mean-squared error, while the Neural ODE achieves the best rollout RMSE at horizons 50, 100, and 500 and the strongest robustness under perturbation. The experiments show that chaotic systems should be judged not only by local prediction error, but also by rollout behavior, attractor structure, and sensitivity to numerical discretization.**

I. Introduction

Scientific Machine Learning (SciML) combines numerical modeling and machine learning for systems governed by scientific structure. The Lorenz equations provide a compact benchmark:

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = xy - \beta z, \tag{1}$$

with $(\sigma, \rho, \beta) = (10, 28, 8/3)$. Writing $s(t) = [x(t), y(t), z(t)]^\top$, the system has the vector-field form $\dot{s}(t) = f(s(t))$. The nonzero equilibria

$$s_\pm = \left(\pm\sqrt{\beta(\rho-1)},\ \pm\sqrt{\beta(\rho-1)},\ \rho - 1\right) \tag{2}$$

are unstable under the standard parameter setting, which contributes to the classical butterfly attractor and the sensitivity to initial conditions. The corresponding discrete flow map is $s_{t+\Delta t} = \Phi_{\Delta t}(s_t)$, so the central learning question is whether a model can approximate either the map $\Phi_{\Delta t}$ or the vector field f while preserving meaningful long-horizon dynamics.

The key challenge is that low one-step error does not guarantee faithful long-horizon behavior. This motivates the three evaluation goals of the project: short-horizon forecasting, rollout stability, and attractor reconstruction. For chaotic systems, even reasonable models diverge pointwise over time, so evaluation must distinguish short-horizon accuracy from qualitative dynamical fidelity.

II. Method

Reference trajectories were generated by fixed-step RK4 integration and split by trajectory into 100 training, 20 validation, and 20 test trajectories. Each trajectory was simulated over $T = 25$ with default step size $\Delta t = 0.01$.

We compare three models. The MLP directly predicts s_{t+1} and serves as the simplest baseline. The residual model predicts an increment and uses

$$s_{t+1} \approx s_t + \Delta s_t, \tag{3}$$

which is consistent with the first-order numerical structure $s_{t+\Delta t} \approx s_t + \Delta t\, f(s_t)$. The Neural ODE instead learns a continuous vector field f_θ and obtains trajectories by integrating $\dot{s} = f_\theta(s)$, making it the model most closely matched to the original continuous-time formulation.

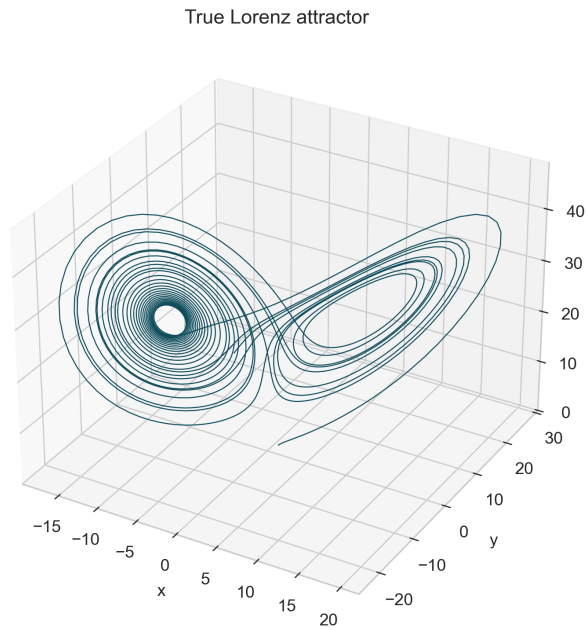

Fig. 1. Reference Lorenz attractor.

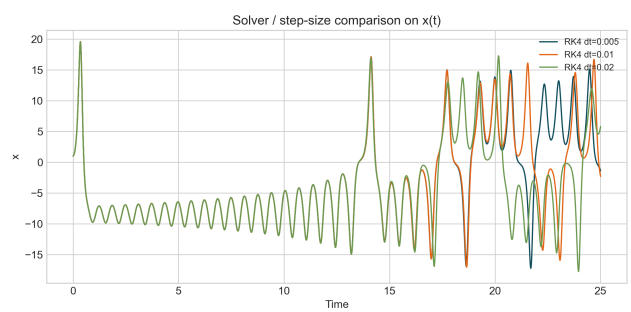

Fig. 2. RK4 step-size comparison on $x(t)$.

We evaluate one-step MSE, rollout RMSE at horizons 50, 100, and 500, attractor-level similarity, and robustness to step-size and noise perturbations. Figure 2 shows that even the same ODE produces visibly different long-time trajectories when the numerical step size changes, which motivates careful simulation and robustness analysis.

III. Results

Table I summarizes the main quantitative results. The residual predictor achieved the best one-step MSE, confirming that increment prediction is an effective local parameterization. However, the Neural ODE achieved the lowest rollout RMSE at all reported horizons, so the best local predictor was not the best dynamical model.

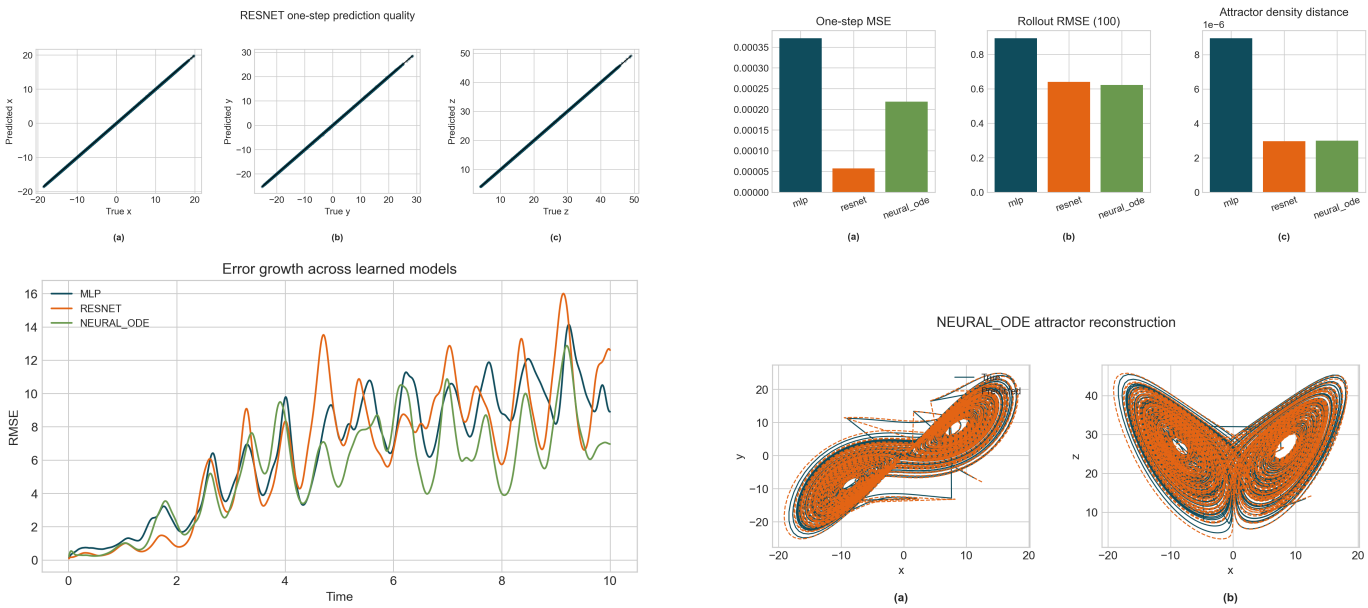

Fig. 3. Top-left: ResNet one-step prediction quality. Top-right: compact overall metric summary. Bottom-left: rollout error growth across models. Bottom-right: Neural ODE attractor reconstruction.

TABLE I
MAIN TEST METRICS.

Model	One-step MSE	RMSE_{50}	RMSE_{100}	RMSE_{500}
MLP	3.72×10^{-4}	0.731	0.894	6.395
Residual	5.71×10^{-5}	0.510	0.641	6.842
Neural ODE	2.18×10^{-4}	**0.438**	**0.622**	**6.011**

The one-step prediction results already show a clear separation between models. The MLP learned accurate one-step transitions, but the residual model improved local accuracy substantially and produced the best one-step metric overall.

Figure 3 summarizes the full story. The top-left panel shows that predicted and true values lie almost exactly on the diagonal for the best one-step model, illustrating how strong short-horizon prediction can look in a chaotic system. The top-right panel places that result in context: ResNet is strongest on one-step prediction, while the Neural ODE becomes more favorable once rollout and dynamical fidelity are considered.

The bottom-left panel makes the short-horizon versus long-horizon distinction clear. The residual model is strongest locally, but the Neural ODE accumulates error more slowly over the evaluated horizon. The bottom-right panel adds the attractor perspective: the Neural ODE preserves the two-lobe geometry of the Lorenz attractor even when pointwise trajectories are no longer identical. Attractor-density comparisons showed the same pattern, and additional robustness experiments under step-size shift and observation noise also favored the Neural ODE.

IV. DISCUSSION

This experiment illustrates a central SciML lesson. For chaotic systems, exact long-horizon tracking is not the only criterion, and often not the right one. What matters is whether the learned model reproduces useful dynamical structure: stable short-rollout behavior, reasonable phase portraits, and attractor geometry.

The residual model is best when the task is framed as discrete next-step regression, whereas the Neural ODE is best when the goal is to respect the underlying continuous-time dynamics over rollout horizons and under perturbation. As summarized in Fig. 3, the Lorenz benchmark shows why SciML should combine numerical simulation, machine learning, and dynamical evaluation rather than relying on one-step prediction alone.